



Leading University

Department of CSE

Lab Assignment No: 03

Course Title: Artificial Intelligence Sessional

Course Code: CSE-4112

Submitted To:

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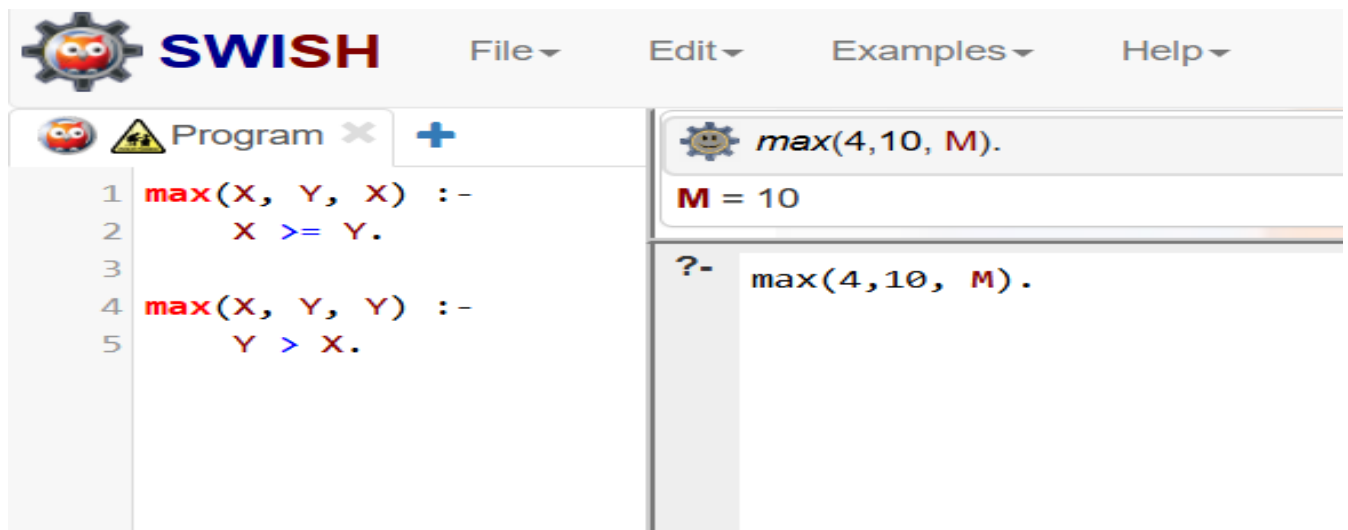
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Solution of Problem 1:

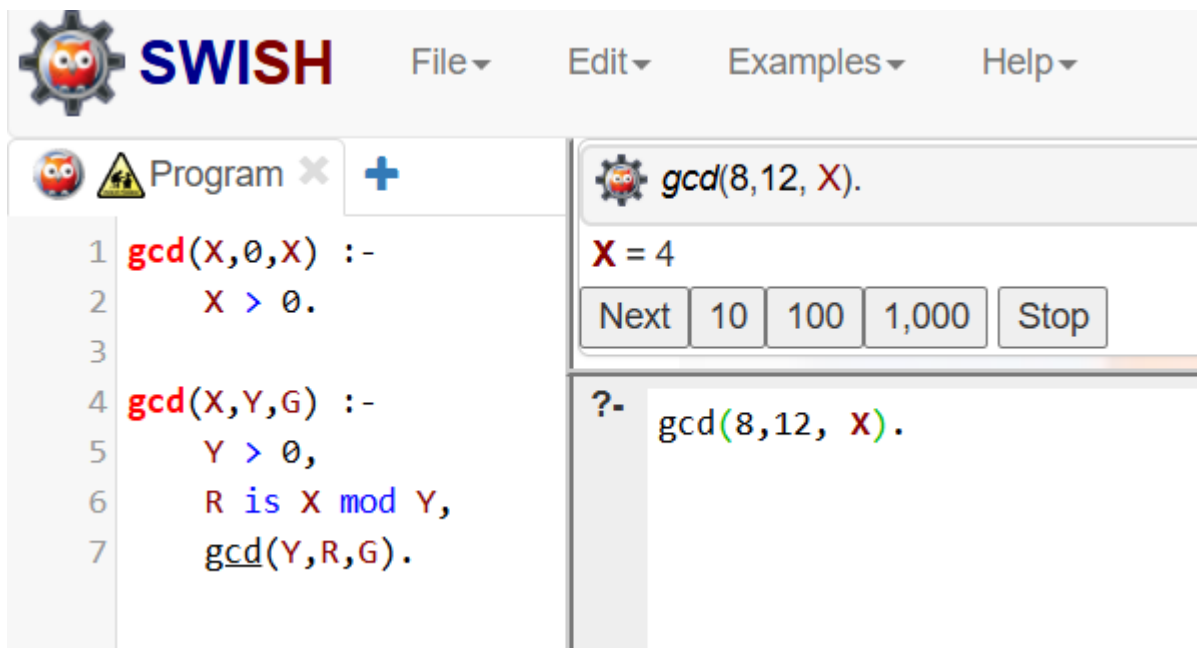


The SWISH Prolog IDE interface shows a program with the following code:

```
1 max(X, Y, X) :-  
2   X >= Y.  
3  
4 max(X, Y, Y) :-  
5   Y > X.
```

The right panel displays the loaded goal `max(4,10, M).` and the binding `M = 10`. Below this, the query `?- max(4,10, M).` is shown.

Solution of Problem 2:



The SWISH Prolog IDE interface shows a program with the following code:

```
1 gcd(X,0,X) :-  
2   X > 0.  
3  
4 gcd(X,Y,G) :-  
5   Y > 0,  
6   R is X mod Y,  
7   gcd(Y,R,G).
```

The right panel displays the loaded goal `gcd(8,12, X).` and the binding `X = 4`. Below this, the query `?- gcd(8,12, X).` is shown. A control bar with buttons for 'Next', '10', '100', '1,000', and 'Stop' is visible.

Solution of Problem 3:

The SWISH interface displays the following Prolog code in the left pane:

```
1 power(_,0,1).
2
3 power(N,P,R) :-
4     P > 0,
5     P1 is P - 1,
6     power(N,P1,R1),
7     R is N * R1.
8
9 start_power :-
10    write('Enter the number : '),
11    read(N),
12
13    write('Enter the power : '),
14    read(P),
15
16    power(N,P,Ans),
17
18    write('Result is : '),
19    write(Ans).
```

The right pane shows the execution of the `start_power` predicate. It prompts for the number and power, with inputs 4 and 2 respectively. The result is 16, and the predicate returns `true`. Below the result, there are buttons for 'Next', '10', '100', '1,000', and 'Stop'.

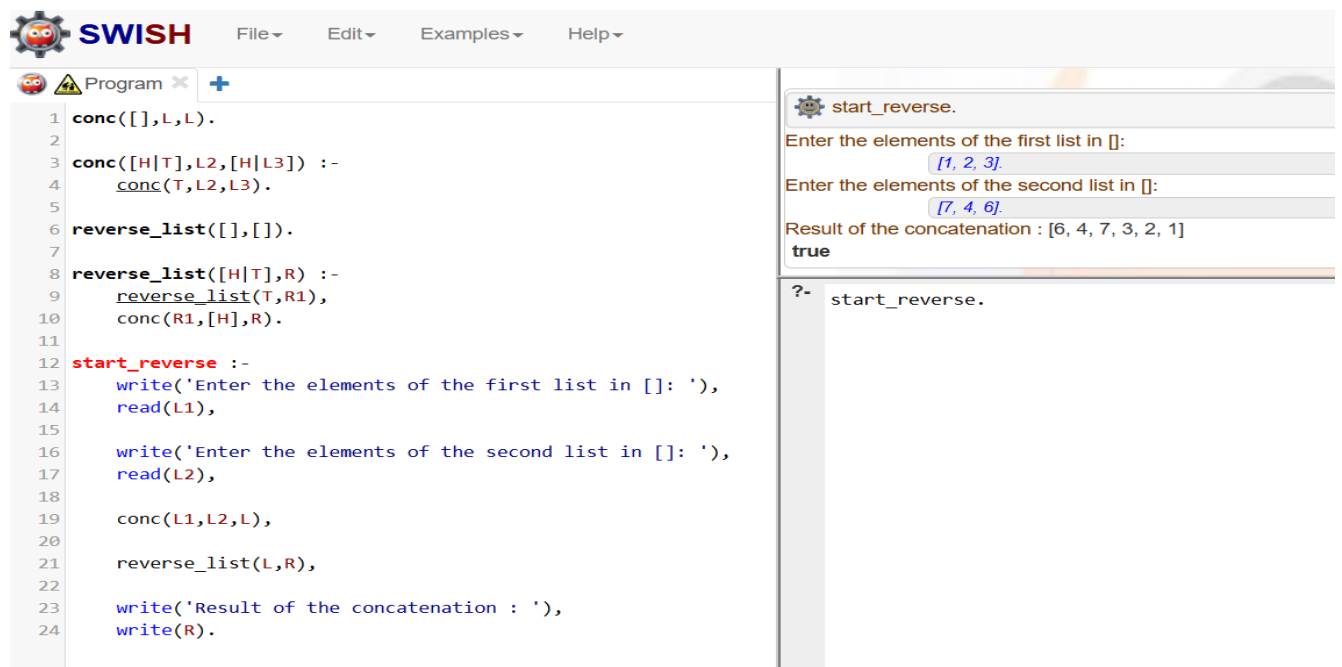
Solution of Problem 4:

The SWISH interface displays the following Prolog code in the left pane:

```
1 conc([],L,L).
2
3 conc([H|T],L2,[H|L3]) :-
4     conc(T,L2,L3).
5
6 start_conc :-
7     write('Enter the elements of the first list in []: '),
8     read(L1),
9
10    write('Enter the elements of the second list in []: '),
11    read(L2),
12
13    conc(L1,L2,L3),
14
15    write('Result of the concatenation : '),
16    write(L3).
```

The right pane shows the execution of the `start_conc` predicate. It prompts for the elements of the first and second lists, with inputs `[1, 2, 3]` and `[7, 4, 6]` respectively. The result of the concatenation is `[1, 2, 3, 7, 4, 6]`, and the predicate returns `true`.

Solution of Problem 5:



SWISH File Edit Examples Help

Program

```
1 conc([],L,L).
2
3 conc([H|T],L2,[H|L3]) :-
4   conc(T,L2,L3).
5
6 reverse_list([],[]).
7
8 reverse_list([H|T],R) :-
9   reverse_list(T,R1),
10  conc(R1,[H],R).
11
12 start_reverse :-
13   write('Enter the elements of the first list in []: '),
14   read(L1),
15
16   write('Enter the elements of the second list in []: '),
17   read(L2),
18
19   conc(L1,L2,L),
20
21   reverse_list(L,R),
22
23   write('Result of the concatenation : '),
24   write(R).
```

start_reverse.

Enter the elements of the first list in []:

[1, 2, 3]

Enter the elements of the second list in []:

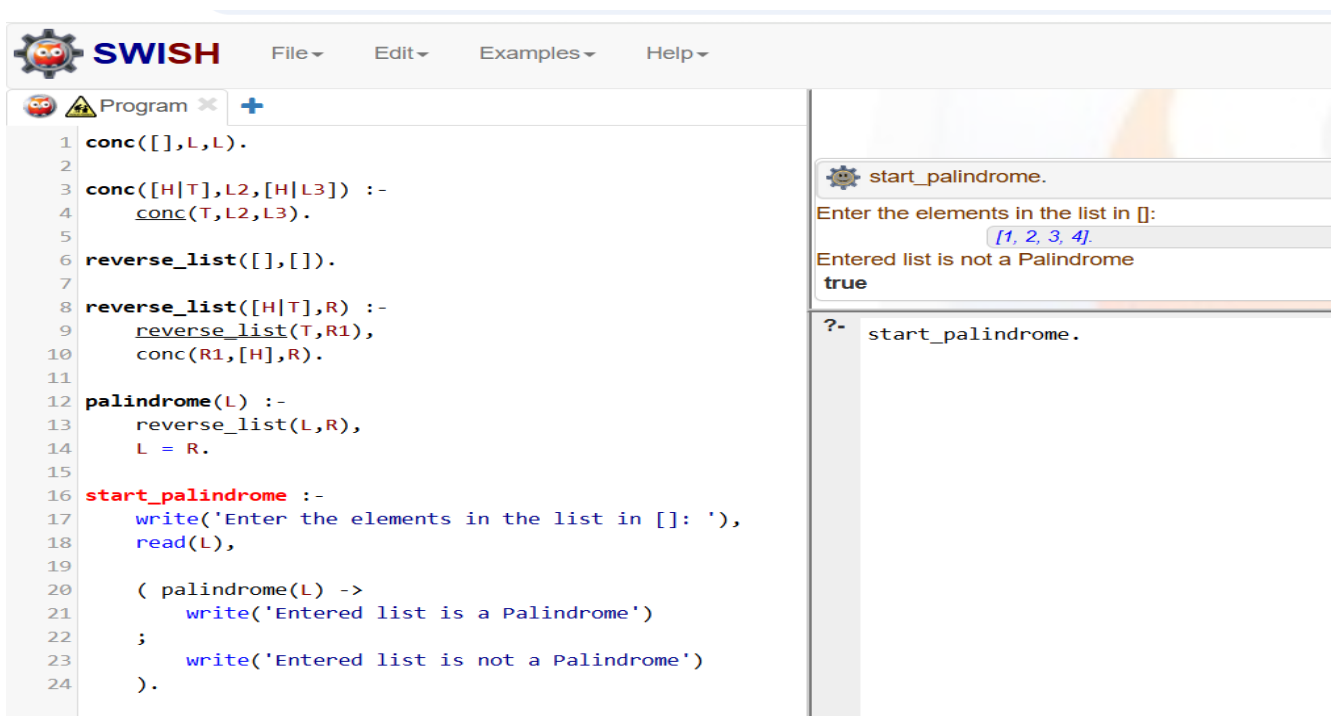
[7, 4, 6]

Result of the concatenation : [6, 4, 7, 3, 2, 1]

true

?- start_reverse.

Solution of Problem 6:



SWISH File Edit Examples Help

Program

```
1 conc([],L,L).
2
3 conc([H|T],L2,[H|L3]) :-
4   conc(T,L2,L3).
5
6 reverse_list([],[]).
7
8 reverse_list([H|T],R) :-
9   reverse_list(T,R1),
10  conc(R1,[H],R).
11
12 palindrome(L) :-
13   reverse_list(L,R),
14   L = R.
15
16 start_palindrome :-
17   write('Enter the elements in the list in []: '),
18   read(L),
19
20   ( palindrome(L) ->
21     write('Entered list is a Palindrome')
22   ;
23     write('Entered list is not a Palindrome')
24   ).
```

start_palindrome.

Enter the elements in the list in []:

[1, 2, 3, 4]

Entered list is not a Palindrome

true

?- start_palindrome.

Solution of Problem 7:

The SWISH Prolog IDE interface shows the following code in the editor:

```
1 memb(X,[X|_]).
2
3 memb(X,[_|T]) :-
4     memb(X,T).
5
6 start_member :-
7     write('Enter element : '),
8     read(X),
9
10    write('Enter list in [] : '),
11    read(L),
12
13    ( memb(X,L) ->
14        write(X),
15        write(' is a member of L')
16    ;
17        write(X),
18        write(' is NOT a member of L')
19    ).
```

The right-hand pane shows the execution of the goal `start_member.` with the following input and output:

Enter element :

Enter list in [] :

3 is a member of L
true

?- start_member.

Solution of Problem 8:

The SWISH Prolog IDE interface shows the following code in the editor:

```
1 edge(s1,a).
2 edge(s1,c).
3 edge(s1,d).
4 edge(s2,c).
5 edge(s2,b).
6 edge(b,e).
7 edge(b,g2).
8 edge(c,d).
9 edge(c,e).
10 edge(e,g1).
11 edge(g1,g2).
12 breadth_first(Start,Goal) :-
13     bfs([Start],[],Goal).
14
15 bfs([],_,_) :- write('Not found').
16
17 bfs([Goal|_],Visited,Goal) :-
18     reverse([Goal|Visited],Path),
19     write(Path).
20
21 bfs([Node|Rest],Visited,Goal) :-
22     findall(X,
23         (edge(Node,X),
24          \+ member(X,Visited),
25          \+ member(X,Rest)),
26         Children),
27     append(Rest,Children,Queue),
28     bfs(Queue,[Node|Visited],Goal).
```

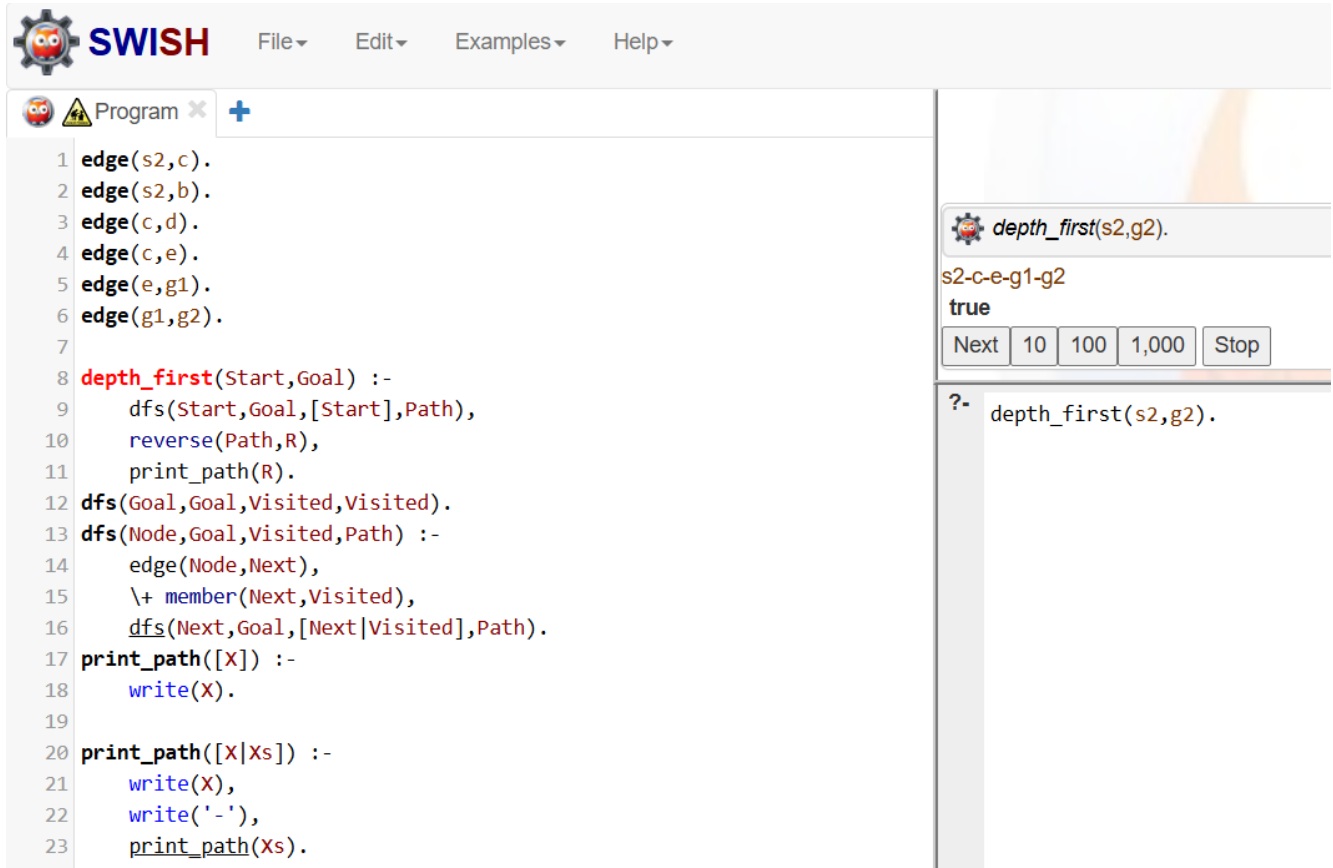
The right-hand pane shows the execution of the goal `breadth_first(s2,g2).` with the following input and output:

[s2, c, b, d, e, g2]
true

Next

?- breadth_first(s2,g2).

Solution of Problem 9:



The image shows the SWISH Prolog IDE interface. The main editor displays a Prolog program with the following code:

```
1 edge(s2,c).
2 edge(s2,b).
3 edge(c,d).
4 edge(c,e).
5 edge(e,g1).
6 edge(g1,g2).
7
8 depth_first(Start,Goal) :-
9     dfs(Start,Goal,[Start],Path),
10    reverse(Path,R),
11    print_path(R).
12 dfs(Goal,Goal,Visited,Visited).
13 dfs(Node,Goal,Visited,Path) :-
14     edge(Node,Next),
15     \+ member(Next,Visited),
16     dfs(Next,Goal,[Next|Visited],Path).
17 print_path([X]) :-
18     write(X).
19
20 print_path([X|Xs]) :-
21     write(X),
22     write(' '),
23     print_path(Xs).
```

The right-hand pane shows the execution results for the query `depth_first(s2,g2).`. The results are:

s2-c-e-g1-g2
true

Below the results, there are buttons for "Next", "10", "100", "1,000", and "Stop".

At the bottom of the right-hand pane, there is a prompt `?- depth_first(s2,g2).`